SNP CrystalBridge

Data Migration
SAP ECC to S/4HANA

Mission Control | Transformation Cockpit | Staging | Near-Zero Downtime | Validation

SCOPE	SOURCE OF CONTENT	SYSTEM SNAPSHOT
SNP CrystalBridge data migration only	Kyano MC Guide, User Guide 25.09, Module 5 v25.03, TC 19.03 decks	Section 2 - landscape template

Prepared: October 2026 | Sempra Migration Team | Version 1.0 (draft - more SNP documents to follow)

How to read this document. Everything marked as a product fact (roles, transactions, reports, parameters, defaults, workflow) is taken from the SNP guide. Values that depend on the Sempra landscape (SIDs, releases, client numbers, volumes, dates) are marked **TBC** and must be completed from the project-specific SNP documents. Nothing in the TBC fields has been assumed.

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01 Executive Overview

What SNP CrystalBridge does for an ECC to S/4HANA data migration

SNP **CrystalBridge** is a suite for planning, executing and validating SAP transformation projects. For a data migration it extracts data from the SAP ECC source, applies selections, mappings and rules, stores the result as compressed export packages and loads it into the S/4HANA target. The source data is **never modified**; all transformation is applied between export and import.

Component	Code	Role in the data migration
CrystalBridge Analysis	AN	Pre-migration analysis: system scan, enterprise structure, data profiling, customizing, repository and interface analysis, S/4 simplification item check.
CrystalBridge Transformation (Transformation Cockpit)	TF / TC	Core migration engine: scenarios, selections, mappings, data objects, rules, export and import blocks, staging, number ranges.
Mission Control (Kyano)	MC	Project plan and orchestration layer: tasks, dependencies, automation of TC steps, tickets, monitoring, templates (units).
Shell Creator	SHC	Creates the system shell (RESC) used as basis for the target in the Bluefield approach.
Data Consistency Verification	DCV	Post-load checks: record counts, foreign key checks, domain fixed values (/N/SNP/FKC).
Rule-Based Near-Zero Downtime	RBNZD	Delta technique so the source stays live and only the last delta is moved at cutover.

Migration approaches supported

Approach	Description	Fit for ECC to S/4HANA
Greenfield	New S/4HANA implementation; processes redesigned; master data and open items only.	Used when history is not required in the target.
Bluefield	Selective data migration per entity into a (global) S/4HANA system; combination of Greenfield and Brownfield. SNP Transformation Backbone with SAP LT.	Selective, harmonizing migration; source can stay live (RBNZD).
Brownfield	Upgrade of the current system to S/4HANA; keeps processes; changes limited to S/4 architecture.	Conversion; target = converted source (no separate target system).

Bluefield unit - supported out of the box (Mission Control): selection by company code; fiscal-year based selection; controlling area / credit control area / operating concern split; chart of accounts harmonization; switch from account to ledger approach, New GL and New Asset Accounting; custom data models 1:1 (Z*/Y*); business partner creation for each customer and vendor.

Needs Professional Services: business partner harmonization, mapping of org units or document types, document splitting, industry solutions, additional currencies, any other transformation requirement.

02 System Snapshot

Architecture, landscape, components, connectivity and run parameters

2.1 Reference architecture

The **control system** hosts Mission Control and the Transformation Cockpit, the rule base and the shadow DDIC. It creates and monitors the export programs on the source and the import programs on the target. Data moves through a compressed file system, never directly between source and target.

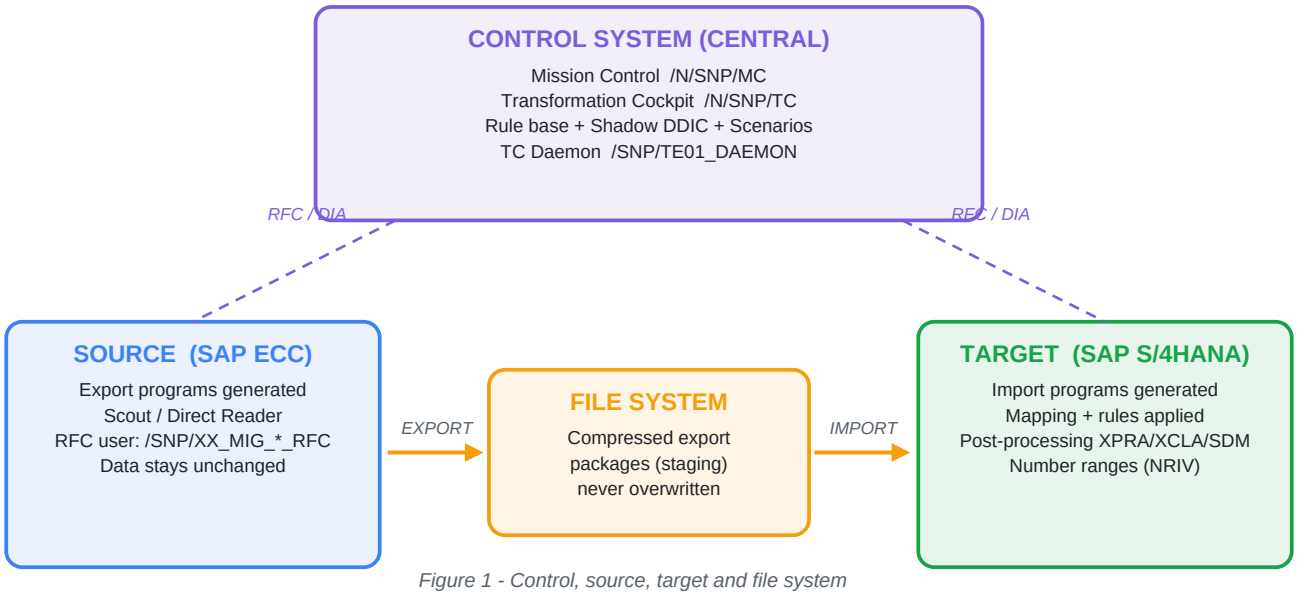


Figure 1 - Control, source, target and file system

2.2 Supported system constellations

Constellation	Description	Notes
Migration, separate control	Control + Source + Target connected via RFC / DIA.	Recommended for large projects.
Conversion, separate control	Control + Source via RFC / DIA; no separate target.	Target = converted source.
No separate control	Source or target acts as control system.	Source recommended; TC slower under load; ruleset export/import needed on refresh (/SNP/TE02_SCENARIO_XML_EXPORT / _IMPORT).
Multiple source systems	Several sources feed one target via one scenario.	Import block must be assigned manually per source; system alias required.

2.3 Landscape snapshot (to be completed)

Fill from the project documents. Each row is a fact the migration team must be able to state at any time.

Attribute	Source (ECC)	Control system	Target (S/4HANA)
System ID (SID)	**TBC**	**TBC**	**TBC**
Client	**TBC**	**TBC**	**TBC**
SAP release / EHP / SP stack	**TBC**	**TBC**	**TBC**
SAP Basis release (drives /SNP/XX_MIG_751 / 755 roles)	**TBC**	**TBC**	**TBC**

Attribute	Source (ECC)	Control system	Target (S/4HANA)
Database system and size	**TBC**	**TBC**	**TBC**
Unicode	**TBC**	**TBC**	**TBC**
Host / IP (for RFC whitelist)	**TBC**	**TBC**	**TBC**
Role in scenario (SOURCE / TARGET / CENTRAL)	SOURCE	CENTRAL	TARGET
Max. process count (parallel jobs)	**TBC**	**TBC**	**TBC**
Export / import file path	**TBC**	**TBC**	**TBC**
SNP software component versions (AN / TF / MC)	**TBC**	**TBC**	**TBC**
SNP transports imported (date / request)	**TBC**	**TBC**	**TBC**
Approach (Greenfield / Bluefield / Brownfield)	**TBC**	**TBC**	**TBC**

2.4 Migration scope snapshot (to be completed)

Item	Value
Company codes in scope (BUKRS)	**TBC**
Controlling area(s) (KOKRS) / Plants (WERKS) / Sales orgs (VKORG) / Purchasing orgs (EKORG)	**TBC**
Fiscal years in scope (GJAHR)	**TBC**
Modules in scope (FI / CO / MM / SD / PP / PM / HCM / other)	**TBC**
Target organizational structure (after mapping)	**TBC**
Custom (Z / Y) tables in scope - count and owner	**TBC**
Estimated data volume (GB) and largest tables	**TBC**
TC scenario ID(s) and type	**TBC**
Number of mock / test cycles planned	**TBC**
Cutover window and downtime target	**TBC**

2.5 RFC connection and whitelist register

All RFC connections used in the transformation must be entered in a whitelist on **all systems and all clients** (maintained in `/SNP/TCSYS`). One row per connection:

Field	Guide example	Project value
System-ID	TRA / SRC / TAR	**TBC**
Client	001 / 800 / 900	**TBC**
IP address	123.45.678.901	**TBC**
RFC destination	SNP_TB_TRA_001	**TBC**
Callback RFC destination	SNP_TB_TRA_001	**TBC**
Dialog RFC destination	SNP_TB_TRA_001_DIA	**TBC**

Connection checks: `Test Connection` in the Systems tab, or Mission Control API reports `/SNP/TE02_API_CHECK_SYS_CONS` and `/SNP/TE02_API_CHECK_FILE_PATHS`.

2.6 Run parameters - defaults and project values

Parameter	Default	Meaning	Project
MAX_PACKAGE_SIZE_MB	50 MB	Max file size per package (override per table)	**TBC**
MAX_PACKAGE_SIZE_REC	200,000	Max records per package file	**TBC**
TASK_PROC_COUNT	4	Processes for other tasks (rule generation, syntax check)	**TBC**
TASK_PACKAGE_SIZE	30	Max objects per process for other tasks	**TBC**
EXEC_PROC_RETRY_CNT	0	Process restarts on rollback / deadlock failure	**TBC**
IN_MEMORY_GENERATION	off	Program generation up to 6x faster; programs not visible in SE38	**TBC**
MAX_IN_MEMORY_PROGRAMS	5	Programs held in memory (range 2-35)	**TBC**
LOG_MAX_ENTRIES	100	Max log entries per table / object	**TBC**
ENCRYPTION_PASS	-	Encrypt personal data; about 30% extra runtime; tables in /SNP/TE00TABENCR	**TBC**
EMAIL_NOTIFICATIONS / EMAIL_INTERVAL_MIN	off / 60	E-mail status dispatch (needs SAP SMTP)	**TBC**
GLOBAL_FUNCTION_EXPORT / _IMPORT	X / X	Generate global functions on source / target	**TBC**
NZD_ACTIVE	off	Enables Near-Zero Downtime delta identification	**TBC**

Prerequisite for any data object: run the **Table Analysis** first. It collects DDIC and record-count information into table `/SNP/CE00TABINFO`. Refresh with report `/SNP/TE02_UPDATE_TABLE_INFO`. Data objects cannot be processed without it.

03 Migration Lifecycle

SNP Bluefield approach and the project phases

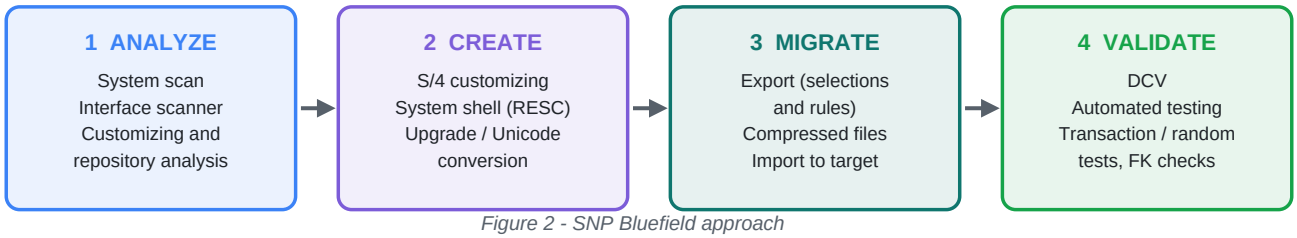


Figure 2 - SNP Bluefield approach

Project phase	Key activities
Project Setup	Present SNP procedure, determine system landscape, project planning, system scan, deliver Transformation Backbone, kick-off.
Analysis	Detailed analysis in CrystalBridge, create data object list, analyze interfaces, customizing and repository analysis, scoping workshops.
Design	Adjust data object list, rule set definition, create blueprint.
Implementation	Rule-set implementation, maintain and check selections and mappings.
Test	Individual tests with protocol, data validation, mass tests, transactional data check, DCV, automated testing.
Go-Live	Go-live simulation, go-live, cutover, data validation, approval.
Hypercare	Support by project team, post-migrations if necessary, lessons learned.

3.1 Analyze - pre-migration analysis tools (CrystalBridge Analysis)

Tool	Transaction / entry	Output used by the migration
System scan (extractor)	/N/SNP/SC	Standard scan (offer) or Extended scan (migration). Produces a .snp file (extension mandatory) loaded to the CrystalBridge platform. Role /SNP/SC required.
Enterprise structure and usage	CrystalBridge platform	Organizational structure, unused company codes / plants (carve-out candidates), usage matrix of data objects vs. domains, module usage.
Master data analysis	/N/SNP/CW_MDA	Compare master data across systems (EQ / NE / MS conflicts), golden-record rule bases, preselection for harmonization.
Customizing analysis	/N/SNP/CW_CUST_COMP, /N/SNP/CW_TAB_MONITOR	Compare customizing tables between systems; create SNP Analysis Cockpit project.
Repository analysis	/N/SNP/BBRAP	Checksum comparison of programs, classes, tables; finds custom code impact.
Interface analysis	CrystalBridge platform	RFC, web service, IDoc and DB connections by volume and frequency.
S/4 Simplification Item Check	Part of system scan	SAP relevance check for S/4HANA.
Master data preselection	/N/SNP/CW_PRESELECTION	Decides which record comes from which source system before export.

Recommended temporary indexes for the scan (remove after success): COBK (MANDT, KOKRS), BKPF (MANDT, GJAHR), VBAK (MANDT, VBTYP), AUFK (MANDT, AUTYP).

04 Transformation Cockpit Configuration

Transaction /N/SNP/TC on the control system

The cockpit is scenario based. A **scenario** is the complete migration blueprint: systems, selections, mappings, data objects, rules and runs. Requires role `/SNP/TF_TC_TC_700_DIA_ALL` (display: `..._DIA_DISP`).

Layer	Object	Description
1	Scenario	Top-level container; source / target systems, selections, mappings, data objects.
2	Transformation run	Named execution plan (Export, Import, Full Load, Delta, Validation, Rollback).
3	Transformation block	Executable unit; processes a set of data objects. Types EXP, FOR, IMP, UPL.
4	Data object	Business entity at table level, including table hierarchy (e.g. FI documents, customer master).

4.1 Scenario types and handling

Type	Use
Bluefield	Selective migration; source stays live.
Greenfield	Full migration to a new S/4HANA system.
Merge	Consolidation of several systems into one target.
Carve-Out	Extract a subset of entities to a new system.

- Create: unique Scenario ID (up to 30 characters); the ID is the value of parameter `P_SCEN` in all Mission Control configs.
- Copy with options: selections, mapping values, data objects, runs and blocks. Delete is irreversible - download first.
- Download: zipped XML named ScenarioName_Date_Time.zip (about 30x size when unzipped). Upload can run in background. Delivery content must be uploaded manually at project start; links to global selections / rules are not validated on upload.
- Scenario variants can be saved locally with `/SNP/HC02_SCENARIO_XML_EXPORT` (ZIP: XML + rule base TXT).

4.2 Selections - scope of the migration

Field	Meaning	Field	Meaning
BUKRS	Company code	VKORG	Sales organization
GJAHR	Fiscal year	EKORG	Purchasing organization
KOKRS	Controlling area	BWKEY	Valuation area
WERKS	Plant	PRCTR / GSBER	Profit center / business area
KUNNR	Customer number	LIFNR / MATNR	Vendor / material number

Range notation (SIGN / OPTION / LOW / HIGH): `IEQ1000` include equal 1000; `ICP100*` include pattern; `IBT10002000` include between; `EEQ9999` exclude equal. Fill global selections from the blueprint with `/SNP/TE02_API_FILL_SEL_FROM_BP`; validate with `/SNP/TE02_VALIDATE_SELECTIONS`.

Selection type	Scope	Reusable	Mode
Global selection	All tables with a field of the assigned domain	Yes	Declarative (AND-linked) or SQL expert
Field selection	Single field in a single table	No	Declarative only

Switching from SQL expert mode back to declarative mode loses all assigned selections. Cached Read (default) loads all values into memory; use Single Read only for memory problems with very large selections.

4.3 Mapping - the transformation layer

Mapping is applied between export and import, in the staging phase. Source data is never changed.

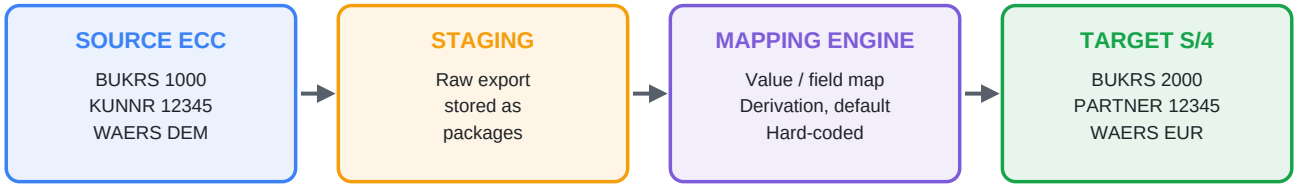


Figure 3 - Mapping applied between export and import

Mapping type	Source keys	Target	Example from guide
1:1	1 value	1 value	WERKS 1000 to 1001
N:1	Several values	1 value	WERKS 1000, 2000 to 2001
1:N	1 value	Several values	WERKS 1000 to 1001 and 2001 - causes an error, values shown in message
X+N:1 (dependent)	2 fields combined	1 value	WERKS + LGORT to LGORT

Mapping layer / method	Description	Example
Global mapping	Applies to all fields of the scope: domain, data element or table field. One rule covers all plant fields in all tables.	BUKRS 1000 to 2000 everywhere
Field mapping	Specific table field only; overrides global; can be deactivated.	BSEG-KOSTL custom logic
Value mapping	Known source value to known target value; wildcards and ranges; CSV bulk import (SOURCE_VALUE;TARGET_VALUE;DESCRIPTION).	WAERS DEM to EUR
Structural field mapping	Source field content to a different target field when the data model differs.	KNA1-KUNNR / LFA1-LIFNR to BUT000-PARTNER
Derivation rule	ABAP logic derives the target from one or more source fields.	PRCTR from KOSTL via CEPC
Hard-coded value	Target always set to a fixed value.	RLDNR = 0L, LOGSYS = target
Default value	Fills only if source field is initial.	GSBER blank to 0001
Concatenate / split / pad / conditional	String and conditional operations.	VKORG/VTWEG/SPART to key

Standard mapping objects

Object	Field	Module	Object	Field	Module
MAP_BUKRS	BUKRS	All	MAP_GSBER	GSBER	FI
MAP_KOKRS	KOKRS	CO	MAP_KOSTL	KOSTL	CO
MAP_EKORG	EKORG	MM	MAP_HKONT	HKONT	FI
MAP_VKORG	VKORG	SD	MAP_WAERS	WAERS	FI
MAP_WERKS	WERKS	MM/PP	MAP_KUNNR	KUNNR	SD
MAP_LGORT	LGORT	MM/WM	MAP_LIFNR	LIFNR	MM
MAP_BWKEY	BWKEY	MM	MAP_MATNR	MATNR	MM
MAP_PRCTR	PRCTR	CO	MAP_ANLN1 / MAP_BLART	ANLN1 / BLART	FI-AA / FI

Object	Field	Module	Object	Field	Module
MAP_MWSKZ	MWSKZ	FI/MM	MAP_ZTERM / MAP_SEGMT	ZTERM / SEGMENT	FI,SD,MM / FI(S4)

Setting up mapping - steps

- Open the scenario, go to the Mapping tab (object catalog).
- Activate required mapping objects (e.g. MAP_BUKRS, MAP_HKONT, MAP_KUNNR).
- Edit values: source value, target value, description; wildcards (*) and ranges supported. More specific entries win over wildcards.
- Bulk upload via Import from File (semicolon CSV). Derivation rules on the Rules sub-tab; hard-coded and default values on their sub-tabs.
- Save, then run `/SNP/TE02_VALIDATE_MAPPINGS` to find gaps. Re-generate blocks (`/SNP/TE02_API_CREATE_BLOCKS`) after major changes.

Best practice: start with org-unit mappings (BUKRS, KOKRS, WERKS, VKORG); use CSV for large GL account tables; validate before every test run; document derivation rules; run in simulation first; store key target values as Mission Control variables; obtain business sign-off on value mappings (role `/SNP/TF_TC_TC_700_DIA_BUSI`).

4.4 Data objects

A data object represents business data at table level, including table hierarchy and dependencies. Add from the catalog (Data Objects tab, Adopt). Custom Z-tables are covered by the **Data Object Builder** (role `/SNP/TF_CO_DOB_700_DIA_ALL`).

Module	Examples of data objects
FI	G/L account master, open items AR/AP, G/L balances, asset master (FI-AA), down payments, recurring entries
CO	Cost center master and actuals, internal orders, profit center master, CO-PA
MM	Material master, purchase orders, scheduling agreements, goods receipts, inventory, info records
SD	Customer master, sales orders, deliveries, billing documents, pricing conditions, credit management
PP / PM / CS / WM	Production orders, BOMs, routings, work centers; functional locations, equipment, maintenance orders; warehouse structures
HCM / ISU / Basis	Employee and payroll; contracts, devices, installations; number ranges, custom Z-tables, application log

Table property	Options / meaning
Export method	SR Scout Reader (FOR ALL ENTRIES, uses last export of selection table) or DR Direct Reader (INNER JOIN, independent).
DKP - duplicate key prevention	Auto-activated when selection table has more key fields than used for the dependent table; slow (no parallelization). Use Direct Reader or adapt structure.
Selection mode	F field selection (declarative) or S free selection (SQL expert).
Package / FAE size / Blocking factor	Override scenario package size for large tables; sub-package segmentation.
Import mode per table	FROM_BLOCK, INSERT, UPDATE, MODIFY, DELETE, NO_IMPORT, FROM_TABLE.

Validate objects with `/SNP/TE02_VALIDATE_OBJECTS`, find missing DDIC with `/SNP/TE02_FIND_MISSING_DDIC`, syntax check with `/SNP/TE02_API_EXEC_SYNTAX_CHK`. Export the object table list to CSV with `/SNP/TE02_TABLES_TO_CSV`. **Address object** (domains ADRNR, AD_ADDRNUM, AD_PERSNUM, AD_SO_KEY, CADRNR): export **last**, import **first**; address numbers are collected in global selection SEL_ADRNR.

4.5 Number ranges (NRIV)

Strategy	Use case
Increment	Internal number ranges in target - shift existing range upward.
Difference calculation	Internal number ranges - calculate gap between source and target.
1:1	Internal number ranges - copy as-is.
Inactive (adopt 1:1)	External number ranges in target.

Functions: Save NRIV, Check NRIV, Update NRIV, Mass Maintenance. Lock number ranges on the target before delta imports and unlock afterwards.

05 Rules, Import Modes and Events

Export rules (consultant) and import rules (migration specialist)

5.1 Import modes

Mode	Existing key	New key
Insert	Terminates with duplicate key error	Inserts record
Insert ADK	Skips record (no error)	Inserts record
Update	Updates non-key fields	Nothing imported
Modify	Updates non-key fields	Inserts record
Delete	Removes record by key	Nothing done

Special care for Insert ADK, Modify, Update and Delete: agree with the customer which records are retained. A field without an import rule is transferred 1:1.

5.2 Export event sequence

Before data object export; process initialization; before table processing; before package processing; before data record (EV_SKIP = 'X' skips a record); global export rule; after data record; after package; global event after package; after table; after data object export. Changes in an earlier event are effective in all later events of the same cycle.

5.3 Rule interfaces

Rule type	Key parameters
Global import rule	IV_VALUE (incoming), EV_VALUE (outgoing), EV_SKIP
Field rule	IS_ (source record), CV_ (target field), EV_SKIP
After data record	IS_ original, CS_ target changeable, EV_SKIP
After package	CT_ package table changeable
Parameter prefixes	I importing, C changing, E exporting; V value, S structure, T table

5.4 API calls available in rules

Function	Purpose
API_MAPPING_ADD / _SET / _GET / _CLEAR	Add (dump if exists) / add or overwrite / read (sy-subrc 4 = 1:1 passthrough) / clear auto-added entries.
API_SELECTION_ADD / _GET / _EXISTS / _CLEAR	Maintain global selections during export (visible on source only; selection must exist and be generated).
API_MESSAGE_ADD	Log message S/E/W/I/A/X; types X and A always written.
API_LOCK / API_UNLOCK	Named lock between parallel processes (visible in SM12).

```
* Global export rule - collect values across packages
* Process initialization
PERFORM api_selection_get USING 'PRCTR' CHANGING lt_prctr.
* Before table processing
SELECT vbeln posnr FROM vbap INTO TABLE lt_vbeln WHERE vbap-prctr IN lt_prctr.
* After package processing
PERFORM api_selection_clear USING 'SEL_VBELN_PRCTR'.
PERFORM api_selection_add   USING 'SEL_VBELN_PRCTR' lw_vbfk-vbeln.
```

Tools: sample feature (F4) generates template code; CTRL+Space auto-completion; version management (compare two versions). Syntax check runs against the shadow DDIC in dialog or background and executes automatically when export or import starts - the transformation terminates on a syntax error. Debugging works in dialog mode only; stop parallel processes first.

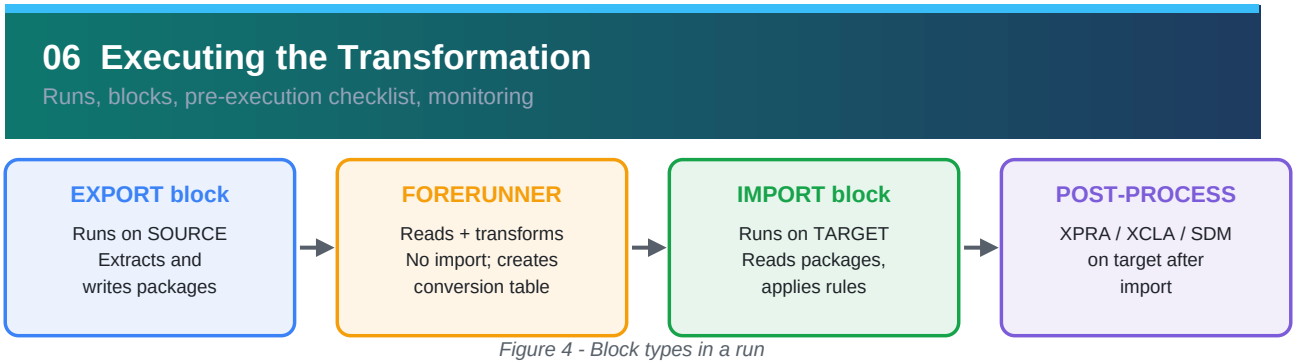


Figure 4 - Block types in a run

6.1 Block configuration

Option	Description
Package size	Records per commit (default 10,000 at block level). Larger = faster but more memory.
Parallel processes	Concurrent jobs; limited by Max. process count of the system.
Restart capability	Restart from last successfully processed package.
Sequence / dependencies	Execution order within the run.
Index suspension	Suspend indexes during import for performance; rebuild afterwards.
Without rule base	Bypass all transformation rules.
Block / table mode	INSERT, INSERT_ADK, UPDATE, MODIFY, DELETE; table level also FROM_BLOCK, NO_IMPORT, FROM_TABLE.
Alias of the system	Required when two or more source systems are assigned.

Create blocks automatically with `/SNP/TE02_API_CREATE_BLOCKS`; copy runs with `/SNP/TE02_API_COPY_RUNS_BLOCKS` when starting a new cycle. Export files are created per run and never overwritten; deleting a block deletes its files; import uses the last export by default (re-assign via context menu).

6.2 Pre-execution checklist

#	Check	Report / action
1	Systems connected, RFC destinations working	/SNP/TE02_API_CHECK_SYS_CONS
2	Selections complete and valid	/SNP/TE02_VALIDATE_SELECTIONS
3	Mappings complete and valid	/SNP/TE02_VALIDATE_MAPPINGS
4	Data objects valid	/SNP/TE02_VALIDATE_OBJECTS
5	DDIC analysis executed	/SNP/TE02_API_REMOTE_DDIC_ANA
6	Syntax check executed	/SNP/TE02_API_EXEC_SYNTAX_CHK
7	Violation check passed	/SNP/TE02_API_VIOLATION_CHECK
8	Target user authorizations sufficient	Roles per section 11
9	Number ranges locked on target (if required)	NRIV area / control block

6.3 Execution options

Option	Use
Background mode	Recommended for production; dialog only for testing / debugging.
Simulation	Validates extraction and transformation logic without writing to the database.
Restart	Resume a failed block from the last successful commit package.
Reset	Reset status and delete intermediate results before re-executing.
Schedule / start immediately	Job priority A/B/C; schedule for a future time.

6.4 Monitoring

Tool	Content
Log tab / /SNP/TCPROT	Start / end, records per object, written to target, errors with table references, records per second; filter options.
Not migrated tables	Exported-versus-imported status per table.
TC Daemon /SNP/TE01_DAEMON	Background SLEEP process (cycle up to 5 s); updates display and drives dependency-based starts.
System Monitor	Remote file system, lock entries, logged-on users, process overview with scenario / run / block, system log.
Remote SE16 /SNP/CN_SE16	View source or target table content without logon.
Run steering	Reset, Refresh, Stop, Restart, Duplicate at run, block, object and table level.

07 Mission Control Automation

Plan, orchestrate, execute and monitor the migration project (/N/SNP/MC)

Mission Control runs on the central system; no additional transports or RFC settings are needed (unlike the cockpit). A project is created from a **unit** (template), must be **licensed** before tasks can run (download license file, support ticket at support.snpgroup.com, upload returned file).

Task type	Description
Manual activity	Executed by a person; status reported manually.
Report call in dialog	Report on a defined system in dialog; status reported manually.
Report call in background	Background job; status derived automatically.
Summary task	Non-executable structure element; variables for all sub-tasks.

7.1 Typical automation configurations

Execute TC export blocks (central system, with automatic ticket creation on error):

```
<config R:CLASS="/SNP/CN02_CL_TEX_REPORT"
        R:REFRESH_CLASS="/SNP/TE02_CL_TEX_TR_REFRESH"
        R:REMOTE_CALLTYPE="LA" R:IV_REPORT="/SNP/TE02_API_EXECUTE_BLOCKS"
        P_SCEN="{M:GET_VARIABLE TC_SCENARIO}" S_RUN="Export" S_BLOCK="BLK_*">
  <R:COMMAND CLASS="/SNP/TE02_CL_CMD_API_GOTO_TC"
             SCENARIO="{M:GET_VARIABLE TC_SCENARIO}" RUN_NAME="Export"/>
</config>
```

Open a selection in the cockpit:

```
<config R:CLASS="/SNP/CN02_CL_TEX_REPORT" R:REMOTE_CALLTYPE="LS"
        R:IV_REPORT="/SNP/TE02_SCENARIO_GOTO"
        P_SCEN="{M:GET_VARIABLE TC_SCENARIO}" P_G_SEL="X" P_ID="BUKRS" P_N_MODE="X"/>
```

Remote SE16 on the source system:

```
<config R:CLASS="/SNP/CN02_CL_TEX_REPORT" R:REMOTE_CALLTYPE="LS"
        R:IV_REPORT="/SNP/CN02_SE16" P_TAB="T001"
        P_RFC="{M:GET_SYSTEM_RFC SOURCE}" R:IV_SHOW_SELECTION_SCREEN="X"/>
```

7.2 Transformation Cockpit API reports used from Mission Control

Group	Report / class	Purpose
Scenario	/SNP/TE02_API_SCEN_FROM_MC	Create TC scenario from Mission Control (P_FILE or P_R_SCEN).
Scenario	/SNP/TE02_API_IMPORT_SCEN_MC	Extend existing scenario with another TC scenario.
Systems	/SNP/TE02_API_SYNC_SYSTEMS	Assign systems to TC scenarios (errors become tickets).
Preparation	/SNP/TE02_API_REMOTE_DDIC_ANA	Remote DDIC analysis.
Preparation	/SNP/TE02_API_FILL_SEL_FROM_BP	Fill global selections from blueprint.
Validation	/SNP/TE02_VALIDATE_SELECTIONS / _MAPPINGS / _OBJECTS	Check selections, mappings, data objects.
Blocks	/SNP/TE02_API_CREATE_BLOCKS / _COPY_RUNS_BLOCKS / _EXECUTE_BLOCKS	Create, copy, execute (P_SCEN, S_RUN, S_BLOCK).
Logging	/SNP/TE02_API_REP_TRANS_PROT / _PROT_FINDINGS	Show migration log; filter log contents.
Proxy	/SNP/TE01_CL_API_TDEF_PROXY	Sync selections and mappings between MC variables and TC.
Proxy	/SNP/TE01_CL_API_MAP_VAL_PROXY / _SCE_PAR_PROXY	Read target mapping values; scenario parameters.
Sync	/SNP/CN02_API_CB_SYNC_PROJECT	Synchronize scenario from blueprint.

7.3 Tickets and forced actions

- Warnings and errors become **tickets**; the task stays **Blocked** until tickets are resolved. Close tickets as FIXED / NO_CHANGE / IGNORE with comment.
- Closing a ticket as 'ignored' sets status Erroneous and may cause issues in later tasks.
- Forced close / reset / execute must be enabled in Project settings (ALLOW_FORCED_CLOSING, Allow forced reset, Force execution); use with caution.
- Task daemon job `/SNP/CN01_TASK_DAEMON` updates asynchronous tasks; process count in registry MC.TASK_DAEMON.PROCESS_COUNT (default 2).
- Projects can be downloaded / uploaded as ZIP (software version must match exactly) and saved as new unit versions.

08 Near-Zero Downtime and Delta Loads

Rule-Based Near-Zero Downtime (RBNZD)

RBNZD lets the source stay live while the bulk of the data is migrated in advance. Only the final delta (changes since the last run) is moved during the cutover window.

Bar height = data volume transferred. Each delta run shrinks the final cutover window.

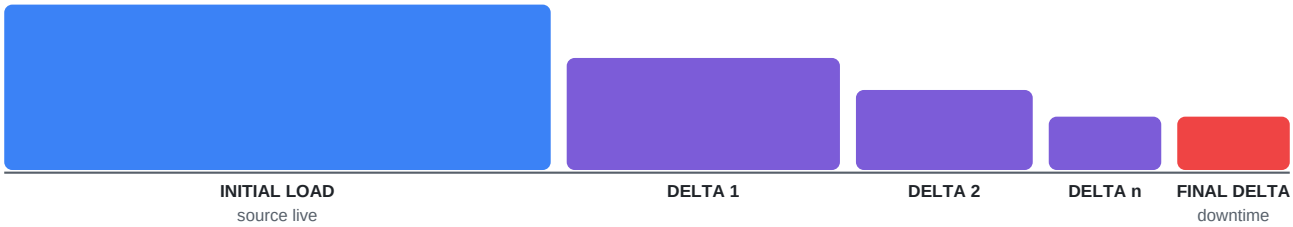


Figure 5 - RBNZD load sequence

Phase	Description
1 Initial load	Full extraction and import while source is live.
2 Delta sync	Multiple delta runs; repeat to shrink the final delta.
3 Final delta	Short downtime: source locked, final sync, target go-live.

8.1 Setting up RBNZD

- Assign role `/SNP/TF_DO_RBNZD_700_DIA_ALL`; set run parameter NZD_ACTIVE.
- Create the scenario with `/SNP/ZD02_TC_CREATE_SCENARIO`; open the hub with `/SNP/ZD02_SCENARIO_GOTO`.
- Configure delta detection rules per data object.
- Execute the initial load through the standard TC blocks.
- Create delta loads with `/SNP/ZD02_MC_CREATE_DELTA_LOAD`.
- Execute with class `/SNP/ZD01_CL_MC_EXECUTE_TASK` from Mission Control; monitor via `/SNP/ZD02_CL_CMD_SCENARIO_GOTO` and refresh class `/SNP/TE02_CL_TEX_TR_REFRESH`.
- Delete a scenario or part of it with `/SNP/ZD02_DELETE_SCENARIO` (errors become tickets).

8.2 Delta process rules

- An initial full load must be completed first; a **delta timestamp** is recorded after each successful export.
- Next delta extracts only records created or changed after the timestamp; import applies in update mode on top of existing data.
- Run deltas frequently (daily or more); validate each delta with DCV before the next.
- Lock number ranges on the target before each delta import and unlock after.
- Track timestamps to avoid overlap or gaps; test the full delta process including rollback in QA before production.

09 Validation and S/4 Post-Processing

Prove completeness and finalize the S/4HANA data model

Check	Tool	What it proves
Compare export to target	Transformation area	Key fields identical; changed non-key fields (mapped values) shown as expected differences.
Data Consistency Verification	/N/SNP/FKC on target; /SNP/FK02_SUMMARY	Domain fixed values (DD07L), foreign key and value-table checks. Supports ECC, HCM, S/4HANA. Results by check, table, domain; before / after comparison.
Balance sheet comparison	RFBILA00 + /SNP/CW02_VR_API_BALANCE_CMP	Financial balances equal between systems; variable VR_IS_MAP_ACTIVE (XFELD) activates TC mapping in comparison.
Parametric transaction comparison	/SNP/CW02_VR_PARAMETRIC_TRX	Same SAP transaction run on both systems; spool outputs compared. Field mapping in variable VR_FIELDS_MAP.
Violation check	/SNP/TE02_API_VIOLATION_CHECK	Referential integrity and mandatory fields.
Automated testing	Role /SNP/TF_AT_700_ALL	Application-level tests via SAP GUI in read and change mode on selected / random test data.
Test units	Cockpit to target	Stored SAP reports and transactions launched from the transformation.
Business Object Detail Analysis	/N/SNP/CW_BODA	HTML report per system for business objects of a TC scenario.

9.1 S/4HANA post-processing

Program	Purpose
XPRA	Post-processing routines adjusting migrated data to the S/4HANA data model.
XCLA	Classification and characteristic value assignments.
SDM (Selective Data Migration)	S/4-specific programs for Finance, Logistics etc.; role /SNP/TF_TC_PP_SAT_754_ALL.

Authorization for post-processing: `/SNP/TF_TC_PP_700_ALL`. Performance optimization role: `/SNP/TF_PF_700_ALL`. The S/4HANA Cockpit uses `/SNP/TF_TC_S4_SAT_751_ALL`.

Business partner creation: ECC customers and vendors are created as business partners in the target. BAPI-based migrations (e.g. business partners) need the additional role /SNP/XX_MIG_BAPI_SAT_700/751/754_ALL on the target. Business partner harmonization is a Professional Services item.

10 End-to-End Workflow to Go-Live

Scenario creation to post-processing (16 steps from the guide)

#	Action	Tool / transaction	Owner	Status
1	Create scenario	/N/SNP/TC	Migration lead	TBC
2	Assign systems, test connections	Systems tab; /SNP/TE02_API_CHECK_SYS_CONS	Basis	TBC
3	Define selections (company code, fiscal year ...)	Selections tab	Functional	TBC
4	Configure mapping	Mapping tab	Functional + business	TBC
5	Add data objects	Data Objects tab	Migration lead	TBC
6	Validate setup	/SNP/TE02_VALIDATE_SELECTIONS / _MAPPINGS / _OBJECTS	Migration lead	TBC
7	DDIC analysis	/SNP/TE02_API_REMOTE_DDIC_ANA	Technical	TBC
8	Syntax check	/SNP/TE02_API_EXEC_SYNTAX_CHK	Technical	TBC
9	Create blocks	/SNP/TE02_API_CREATE_BLOCKS	Technical	TBC
10	Test execution (simulation)	Execute blocks in simulation mode	Technical	TBC
11	Test migration (QA)	Full load into QA target	Migration team	TBC
12	Validate results	/SNP/FK02_SUMMARY; /SNP/CW02_VR_API_BALANCE_CMP	Functional	TBC
13	Production initial load	Export + Import blocks in background	Migration team	TBC
14	Delta runs (RBNZD)	/SNP/ZD01_CL_MC_EXECUTE_TASK (repeat)	Migration team	TBC
15	Final cutover delta	Lock source, final delta, validate, go-live	Cutover manager	TBC
16	Post-processing	XPRA / XCLA / SDM	Technical	TBC

Owners shown are proposed role types, not names from the SNP guide; confirm against the Sempra RACI. Status column is for use during the project.

11 Roles and Authorizations

Composite and single roles for dialog and RFC users

Composite roles reduce effort. They do **not** contain roles delivered separately, such as `/SNP/XX_MIG_755_RFC` or `/SNP/TF_TC_PP_SAT_754_ALL`, which must be installed and assigned separately.

Role	Purpose	System
/SNP/TF_DIA_ALL	Composite - dialog users for TF and AN	Control + satellites
/SNP/TF_RFC	Composite - RFC users for TF and AN	Control + satellites
/SNP/TF_TC_TC_700_DIA_ALL	Transformation Cockpit (full)	Control
/SNP/TF_TC_TC_700_DIA_BUSI	Business users maintain selections and mapping values	Control
/SNP/TF_TC_TC_700_DIA_DISP	Read access to scenarios	Control
/SNP/MC_700_DIA_ALL	Mission Control	Control
/SNP/AN_700_DIA_ALL	CrystalBridge Analysis	Control
/SNP/XX_MIG_700_RFC (+751, 755)	Cross-component RFC user; 755 for generating tables on Basis 7.55 (or auth object S_ABPLNGVS ABP_LNG_VS='', ACTVT 40)	Source, target
/SNP/XX_MIG_BAPI_SAT_700/751/754_ALL	Migrations using BAPIs (e.g. business partners)	Target
/SNP/TF_TC_SG_700_DIA_ALL / _DISP / _RFC	Staging framework (use / review logs / cross-system RFC)	Control, satellites
/SNP/TF_DC_DCV_700_DIA_ALL / _DISP / _RFC	Data Consistency Verification (use / display results / RFC)	Control, satellites
/SNP/TF_DC_DCV_751_DIA_ALL	DCV - Foreign Key Finder	Control
/SNP/TF_DO_RBNZD_700_DIA_ALL	Rule-Based Near-Zero Downtime	Control
/SNP/TF_TC_S4_SAT_751_ALL	SNP S/4HANA Cockpit	Target
/SNP/TF_TC_PP_700_ALL	S/4 post-processing (XPRA, XCLA, SDM)	Control, target
/SNP/TF_TC_PP_SAT_754_ALL	S/4 post-processing, SDM-specific (separate install)	Target
/SNP/TF_PF_700_ALL	Performance optimizations	Control, satellites
/SNP/TF_CO_DOB_700_DIA_ALL	Data Object Builder	Control
/SNP/TF_AT_700_ALL	Automated Testing	All
/SNP/TF_TC_AM	Archive Migration	Target
/SNP/SC	CrystalBridge Analysis extractor (system scan)	Scanned system

Authorization profile	Description
/SNP/ADMINISTRATION	Full authorization in all cockpits
/SNP/CE_ADMIN	Full authorization in the Transformation Cockpit
/SNP/CE_SCENARIO_IMPL	Maintain rules
/SNP/USER, /SNP/PROJECTMANAGER	Standard user / project manager
/SNP/CW_DEVELOPER, /SNP/CW_DISPATCHER	SNP Analysis Cockpit developer / dispatcher

12 Troubleshooting, Risks and Open Items

Known issues from the guide and items to confirm

12.1 Known issues and resolutions

Issue	Symptom	Resolution
Unmapped source value	Import fails: no target value found for BUKRS=1200	Add entry to MAP_BUKRS; run /SNP/TE02_VALIDATE_MAPPINGS to find all gaps.
Mapping not applied	Source value unchanged in target	Check object is activated and block includes the relevant tables.
Stale mapping cache	New entries not in test run	Re-generate blocks via /SNP/TE02_API_CREATE_BLOCKS.
Wildcard conflict	Wrong value mapped	More specific entries win; review mapping table sequence.
Case sensitivity	EUR does not match eur	Values are case-sensitive; use consistent casing.
Circular mapping	Endless loop in transformation	Check derivation classes for recursion; run syntax check.
Data object not processable	Processing refused	Run Table Analysis (/SNP/TE02_UPDATE_TABLE_INFO).
Slow export with DKP	Low throughput	Use Direct Reader or adapt object structure.
Block failed	Status Crashed / Canceled	Review log; Restart from last good package; Reset only if re-extraction is intended.
Task blocked	Task cannot be closed	Resolve or properly close tickets in Mission Control.
Debugging parallel runs	Debugger does not stop	Debug in dialog with one process; stop parallel processes.

12.2 Risk register (derived from guide behaviour)

Risk	Impact	Mitigation
Incomplete value mappings	Import errors, rework in cutover	Validate mappings before every cycle; business sign-off.
Import modes Modify / Update / Delete misused	Unintended overwrite or loss in target	Agree retained records; restrict to named blocks.
Delta timestamp gaps or overlaps	Missing or duplicated records	Track timestamps; DCV after each delta.
Number range clashes	Duplicate keys, wrong ranges	NRIV strategy per object; lock before delta.
RFC whitelist incomplete	Connection failures at execution	Maintain /SNP/TCSYS on all systems and clients; test connection.
Role gaps (755 / BAPI / PP SAT)	Generation or import failures	Install separately delivered roles; check section 11.
Software version mismatch on project upload	Project cannot be loaded	Align SNP versions between systems.
No license on new MC project	Tasks cannot be executed	Request license via SNP support before first execution.

12.3 Open items - to be filled from the next SNP documents

#	Open item	Needed from	Status
1	Landscape snapshot values (section 2.3)	SNP system documentation	TBC
2	Scope: company codes, plants, fiscal years, modules (section 2.4)	Blueprint / scoping	TBC
3	Scenario IDs and run / block naming convention	TC scenario export	TBC
4	Mapping specification (org units, GL accounts, document types)	Functional design	TBC
5	Cycle plan: mock loads, volumes, durations	Project plan	TBC
6	Cutover schedule and downtime target	Cutover plan	TBC
7	Business partner approach (CVI / harmonization)	Data design	TBC
8	RACI and named owners for workflow steps	Project governance	TBC
9	Reconciliation tolerances and sign-off criteria	Finance / business	TBC

13 MC / TC Interlink and Options

How Mission Control drives the cockpit, export / import options, one object end to end

13.1 How Mission Control and the Transformation Cockpit are interlinked

Mission Control (`/N/SNP/MC`) plans and orchestrates; the Transformation Cockpit (`/N/SNP/TC`) holds the scenario and moves the data. They run on the same control system and are coupled by **configuration**, not by a shared table: an MC task carries an automation XML that calls a TC API report, and the scenario ID is passed through the project variable `TC_SCENARIO` (report parameter `P_SCEN`).

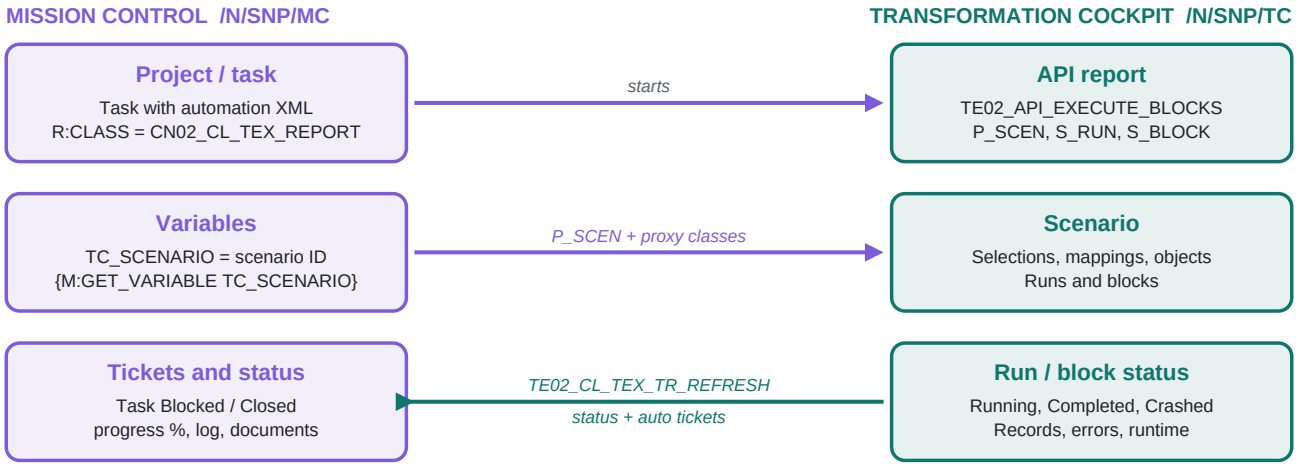


Figure 6 - Mission Control to Transformation Cockpit interlink

Link point	How it works	Where it is configured
Scenario ID	Variable TC_SCENARIO holds the scenario ID and is read in every task with {M:GET_VARIABLE TC_SCENARIO}.	MC project / summary task variables
Start	Report class CN02_CL_TEX_REPORT calls API report TE02_API_EXECUTE_BLOCKS with P_SCEN, S_RUN and S_BLOCK.	Task automation XML
Status back	Refresh class TE02_CL_TEX_TR_REFRESH reads run / block status into the task: progress, log, auto ticket on error.	Task automation XML

Link point	How it works	Where it is configured
Navigation	Command class TE02_CL_CMD_API_GOTO_TC and report TE02_SCENARIO_GOTO open the cockpit on the right scenario or selection.	Task automation XML
Remote checks	CN02_SE16 and similar reports let the task read source or target tables via the RFC of the system alias.	Task automation XML

Backend tables involved

Table	Area	Purpose (from the SNP guides)
/SNP/CN01DAEMLOG	MC	Daemon log of Mission Control background processing.
/SNP/CN00REGUV	MC / TC	Registry parameter table (for example the TE.DAEMON.* settings).
/SNP/CE00TABINFO	Analysis	Output of the table analysis (size, growth).
/SNP/TE00TRPROT	TC	Transformation protocol; can be rebuilt with /SNP/TE02_REBUILD_TRANS_PROT.
/SNP/TE00NRIV	TC	Saved number-range levels on the target (SCEN_ID, OBJECT, NRRANGENR, FROMNUMBER, TONUMBER, NRLEVEL).
/SNP/TE00TABREFT	TC	Table statistics, refreshed by the daemon (interval TE.DAEMON.TAB_STAT_REF_INTERVAL_IN_MIN, default 30 min).
/SNP/TE00TABENCR	TC	List of encrypted tables (HR tables encrypted by default under scenario *).
/SNP/TE05DKP, TE05ESEL, TE05ESELC	TC	Duplicate-key prevention and extended selection tables; cleaned by /SNP/TE05_CLEANUP.
/SNP/FK01FKEYDEF	DCV	Customizing of additional foreign keys for the foreign key check.
/SNP/CW00TABCMP, /SNP/CW00VRCFG*	Validation	Table comparison results and validation report configuration.
/SNP/PE01STGHEAD, PE01STGREG, SG05DDTSTAT, SG05DDDET	Staging	Posting Engine / staging header, registry and status.
/SNP/US00STAT	Support	Usage statistics.

No table that links an MC task to a TC scenario is named in the SNP documents. The link is the scenario-ID variable plus the API report parameters. To see the stored task configuration, search the control system with SE11 / SE16N for `/SNP/CN0*`, `/SNP/CE0*` and `/SNP/TE0*`.

13.2 Export and import inside the Transformation Cockpit

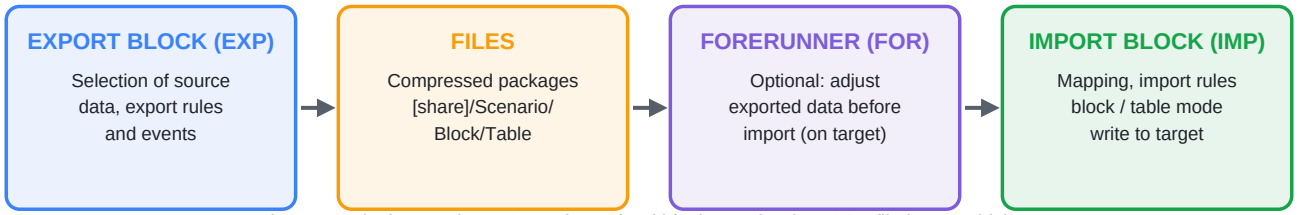


Figure 9 - Block types in a TC run (an upload block UPL loads a CSV file into a table)

Export options

Option	Behaviour	When to use
Scout Reader (SR) - default	FOR ALL ENTRIES. Max 1 process per table; dependent tables run after the selection table using its last export. Restart resets the whole table.	Standard choice for object hierarchies.
Direct Reader (DR)	INNER JOIN. DR tables are start tables, run with exactly 1 process, restart means full re-export. Not usable with concatenated field connections; export code changing join fields is ignored.	Very large dependent tables where the join is cheaper than FAE.
Duplicate key prevention (DKP)	Active when not all key fields of the selection table are used in the connection; forces 1 process per table. A unique selection on the missing key field (for example WERKS = 1000 on MARC / MARA) deactivates it.	Leave on by default; deactivate only with a unique selection.
Selection mode F / S / N	F = field selection; S = free SQL; N = native SQL (set manually).	F for normal scope; S / N for complex conditions.
Blocking factor (FAE)	Default 10 for transparent tables and 5 for pool / cluster tables; FAE package size is separate.	Tune when FAE statements are slow.
Package size	In records or MB; 0 puts all records into one package.	Small packages for restartability; large for fewer files.
Parallel hint (Oracle)	Default 4; applies to start tables and DR tables only.	Oracle sources only.
Stick to same process / Ignore redirect	Keeps a table on one process; ignore redirect generates a proxy table.	Special cases only.
Field vs. global selection	A field selection with no value exports the whole table; a global selection with no value transfers no records.	Check empty selections before every export.

Export files are stored at `[share]/ScenarioID/Block PO ID/Table PO ID/` and are never overwritten. A restart exports only failed tables; child tables of a failed parent are not processed. Related tools: `/SNP/TE02_CHECK_EXPORT_FILES`, browse / download export files, display number of entries.

Import options

Option	Behaviour	When to use
Block mode	INSERT, INSERT_ADK, UPDATE, MODIFY, DELETE. MODIFY or DELETE at block level override the table settings.	INSERT for a clean load; others only by agreement.
Table mode	FROM_BLOCK, INSERT, UPDATE, MODIFY, INSERT_ADK, DELETE, NO_IMPORT.	Exceptions per table, NO_IMPORT to skip a table.
Several import modes	Adds field TE_IMPORT_MODE that rules can set per record, for example cs_mara-te__import_mode = 'MODIFY'.	Mixed new / existing records in one table.
Import option APPEND / SUM	SUM works only with the rule base and numeric non-key fields.	Merging balances into existing records.
With / without rule base	Applies import rules or loads the exported data as is.	Without rule base for pure copies.
Import sequence	1-9999, default 10; lower numbers first.	Header before item tables.

Option	Behaviour	When to use
Select export	If none is selected, the last successful export is used.	Pick an older export to reload.
Differing target table	Imports into a table with another name.	Staging or redirect targets.
Simulation mode	Writes no data; checks duplicates within the package and keys already in the DB; max 100 errors per table and process; not possible once a real import has started.	Before the first real import of a block.
Deletion block	Can be created for an incomplete import block.	Clean-up after a failed import.

Import process distribution avoids lock problems; tables with table locks get priority, and a duplicate-key check runs within each package.

Event order

Run	Order of events
Export	Before object, process init, before table, before package, before record, global export rules, after record, after package, global event after package, after table, after object.
Import	Process init, before table, before package, before record, field rule (overrides the global rule), global import rules / offsets, after record, after package, global event, after table.

Setting `ev_skip = abap_true` skips a record. Global rules are scoped by domain, data element or table field and take up to 20 parameters; offsets handle concatenated fields.

13.3 One object end to end - sales documents (illustrative)

> Illustrative example. The tables, values and names below are chosen to show the mechanics; they are not Sempra values and not taken from a Sempra scenario. Real selection fields come from the SNP data object definition (**TBC**).

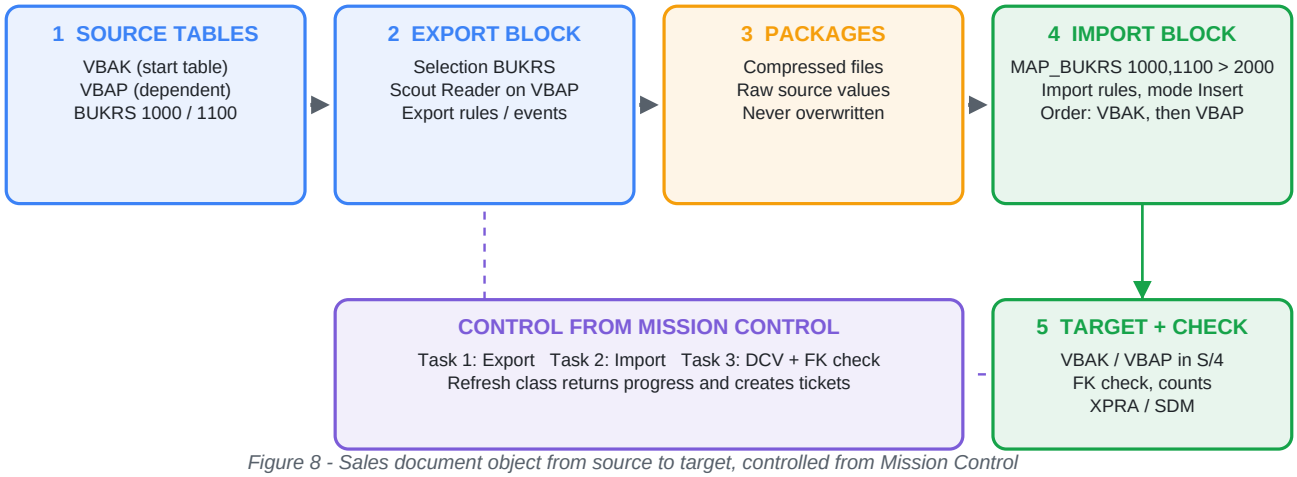


Figure 8 - Sales document object from source to target, controlled from Mission Control

#	System	Action	Backend / API
1	MC	Project from the S4B template; variable TC_SCENARIO set to the scenario ID.	MC project, variables
2	TC	Scenario created, via the Scenario Configurator from an SNP_MIG* template.	/SNP/TE02_SCEN_CONFIGURATOR
3	TC	Global selection on organizational units (example: company codes 1000, 1100).	Selections
4	TC	Mapping MAP_BUKRS: 1000 and 1100 to 2000 (example values).	Mapping objects
5	TC	Data object for sales documents assigned to an export block; start table VBAK, dependent VBAP.	Data object, block
6	TC	Import block created: mode INSERT, import sequence VBAK before VBAP.	Import block
7	TC	Validate objects and syntax check of rules.	/SNP/TE02_VALIDATE_OBJECTS
8	MC	Task 1 starts the export (S_RUN = Export).	TE02_API_EXECUTE_BLOCKS
9	TC	Scout Reader selects VBAK, then VBAP via FOR ALL ENTRIES on the VBAK export.	Export block
10	Files	Compressed packages written; raw source values, never overwritten.	[share]/Scenario/Block/Table
11	MC	Refresh class returns progress; an error creates a ticket and blocks the task.	TE02_CL_TEX_TR_REFRESH
12	MC	Task 2 starts the import (S_RUN = Import).	TE02_API_EXECUTE_BLOCKS
13	TC	Mapping applied between export and import; import rules run; VBAK then VBAP written.	Import block
14	TC	Number ranges saved / checked on the target.	/SNP/TE00NRIV
15	MC	Task 3 runs DCV and the foreign key check; results close the task.	DCV, /SNP/FK01FKEYDEF

Task 1 as automation XML (same pattern as section 7.1; the export block name is illustrative):

```
<config R:CLASS="/SNP/CN02_CL_TEX_REPORT"
        R:REFRESH_CLASS="/SNP/TE02_CL_TEX_TR_REFRESH"
        R:REMOTE_CALLTYPE="LA" R:IV_REPORT="/SNP/TE02_API_EXECUTE_BLOCKS"
        P_SCEN="{M:GET_VARIABLE TC_SCENARIO}" S_RUN="Export" S_BLOCK="BLK_SALES_EXP"/>
```

13.4 When to use what

Situation	Use	Reason
Normal parent / child object	Scout Reader, mode INSERT	Default behaviour, restartable per table.
Huge dependent table, join is cheaper	Direct Reader	INNER JOIN; accepts 1 process and full re-export on restart.
Fix exported data before import	Forerunner block	Adjusts exported data on the target without re-export.
Load a list from Excel / CSV	Upload block	CSV template is downloadable.
Test an import safely	Simulation mode	No data written; duplicate and key checks.
Changed values per record	Several import modes	Rule sets TE_IMPORT_MODE per record.
Large volume, short downtime	RBNZD delta loads	Triggers and collector tables capture changes; see section 8.
Check data without writing to the target	Staging manager	Type D (DDIC, constant runtime) or C (client, linear runtime).

13.5 Additional detail from the Transformation User Guide (25.09)

Topic	Fact
Setup	System maintenance /N/SNP/TCSYS; RFC whitelist /N/SNP/PXRFCWL per client on all systems (use * for the namespace). Template runs must be converted to executable runs.
Scenario Configurator	5 steps: add file, scoping, target system, import preview, cleanup. Parameters include TGT_SYS_CREA_TYPE_TC (N, O, S), SCENARIO_TYPE, TGT_SAP_APPL_VERSN, TGT_S4CORE_VERSN, SC_MANDATORY.
Forerunner (FOR)	Runs on the target, uses only successfully exported FOR-type objects, writes to the target.
Staging	Manager type D (DDIC): shadow tables switched by renaming; pool / cluster tables become transparent. Type C (client): clients AAA-ZZZ. Use cases: downtime minimization, recycle bin, FK check without writing, transformation log with undo.
RBNZD internals	DB triggers write to trigger tables; collector tables and collector programs are generated per root table. Delta loads use the collector tables as additional selection with the same rule base (delete / insert principle). Triggers must be on before the initial load; only TC is supported as engine; non-compliant objects go to the final delta or initial load.
Validation in MC	Tasks: define parameters, set SNP layout as default, run report in source and target, compare (auto or manual), set customer layout as default. The S4B unit template contains them.

14 Advanced TC Techniques

Additional detail from Module 5 (v25.03), SNP2[TC] and SNP3[TC] (TC 19.03)

This section adds what the SNP training decks contain beyond the reference guides. Source tags: **M5** = Module 5 Transformation Advanced Carve-Out v25.03; **SNP2** = SNP2[TC] Transformation Cockpit 19.03 SP00; **SNP3** = SNP3[TC] Advanced Learners 19.03 SP00. The 19.03 decks are older than the 25.09 user guide, so check a point against the 25.09 guide before relying on it for release-specific behaviour.

14.1 Technical process and scenario handling

Topic	Fact	Source
Technical process	The controller on the control system creates and monitors worklists. Export programs on the source read the DB and write files; import programs on the target read the files and write the DB. Both are generated from the rule base and the shadow DDIC. A scenario can be switched from TC to the SAP LT migration workbench (MWB) at scenario level.	SNP2
Skills needed	ABAP, SQL and module experience.	SNP2
Test vs. productive runs	Satellite systems are typically system copies; the file system sits between source and target.	SNP2
Scenario content	Delivery content must be uploaded manually at project start. Scenarios can be created, copied, downloaded (zipped XML named Scenario_Date_Time) and uploaded. When single objects are uploaded, links to global selections, rules, offsets and functions are not validated.	SNP2
Table analysis	Collects DDIC and record counts from source and target into the control system (spool and /SNP/CE00TABINFO). Data objects cannot be processed without it. Refresh with /SNP/TE02_UPDATE_TABLE_INFO.	SNP2
Active tables to CSV	Report /SNP/TE02_TABLES_TO_CSV exports all active tables of a data object.	SNP2
In-memory generation	Programs/source texts are kept in /SNP/PX0XMEMREP and loaded at run time: generation can be parallelized (up to 1/6 of the time) but programs cannot be seen in SE38. Standard version creates reports on the satellite (visible and debuggable, only partly parallel).	SNP2
Training scenario caution	The training project differs from typical project content and procedure.	M5

14.2 Data object design

Mass maintenance of a data object shows these columns per table: number, table name, alias, number of selection table, selection tables (generated path), differing target, status (active / inactive), export predecessor, export method (SR / DR), import sequence (10, 15, 20 ...), mode, import option (APPEND / SUM), sum fields, restrictions and categories (to enable or disable tables). Aliases are needed when a table is used more than once in an object and must be unique inside the object. Inactive tables are ignored in export and import. (M5, SNP2)

Technique	How it works / when to use	Source
Selection vs. dependent table	Header tables drive the records exported; dependent tables hold the details (for example VBAK to VBAP, KNB1 to KNA1). A right-click unlinks a dependent table back to a selection table.	M5
Table connection by key fields	Connect on key fields or on parts of a field (offset and length, for example OBJNR(2)). Free selection mode S allows SQL such as KNA1~KUNNR = {seltab:KUNNR}.	M5, SNP2
Concatenated fields	One field built from several (for example STXH-TDNAME = MATNR + VKORG + VTWEG with TDOBJECT = 'MVKE'). A table extension is filled from the header table; in SQL expert mode the user must set up and fill the extension.	M5, SNP2
Avoiding DKP	Three options: use the Direct Reader, adapt the object structure (table hierarchy, collecting values into a global selection and using it with an export predecessor), or complete the table connection. DKP is slow because it cannot run in parallel and logs every exported key.	M5, SNP2

Technique	How it works / when to use	Source
Copy data objects	Duplicate (1:1 in the same scenario) or copy tables from another object / scenario. Cross-scenario copy transfers global selections and existing rules are not overwritten.	M5, SNP2
Domain usage	Finds table fields using a chosen domain and shows whether the table is already in the scenario (missing, inactive, active); can create a new object. Domain usage with content also checks real values and cross-client tables.	M5, SNP2
Deactivate empty tables	Deactivates tables with no records; must be repeated for each test because contents change. No longer works with multiple source systems (only default systems are checked).	M5, SNP2
View as source	A DB view can be used like a table. Without a differing target table only import mode NO_IMPORT is allowed; with a differing target (for example an own table modelled on the view) data can be imported.	SNP3
CSV upload	Template CSV downloaded, filled by the customer, uploaded. Export methods: upload into an available SAP table, or into a virtual $ table that takes the CSV format.	SNP3
Differing target table	Must exist in the target DDIC. Same-named fields are assigned automatically; unassigned field contents are lost. Fields can be filled with MOVE or by coding; data consistency is the consultant's responsibility.	SNP3
Objects without DDIC reference	A check lists exceptions per system; the affected object can be deactivated.	SNP3

14.3 Selections, own tables, events and functions

Topic	Fact	Source
Selection modes	Field selection (declarative, default, recommended): connection AND field selection AND global selections, always with AND. Free selection (imperative, SQL expert mode): OR / NOT and other combinations in Open SQL. Switching back to declarative loses the assigned selections.	M5, SNP2
Read modes	Cached read (default): all values loaded on first read, fast but can run out of memory on large tables. Single read: values read individually, slow, no complex selections; use when memory is a problem.	SNP2
Global selections	Reusable for every field of the assigned domain (or same type if functionally correct). Values in database format incl. leading zeros. Created centrally in Global or on the table; mass assignment via Functions - Apply global selections (not possible if a selection is already assigned). Deleting a global selection deletes its links. Overview: /SNP/TE02_SELECTION_OVERVIEW.	M5, SNP2
Runtime-filled selections	Initialize with API_SELECTION_CLEAR (event before object export), collect with API_SELECTION_ADD, test with API_SELECTION_EXISTS. API_SELECTION_CLEAR_SCOPED limits clearing to objects and tables. Global selections are generated as sub-selects, not INNER JOINs. Entries are added with MODIFY, are visible on the source only and are shown via the table in source system (remote SE16).	M5, SNP3
Own tables	Namespace /SNP/TE_*; created for source and/or target; any structure; usable like DDIC tables in objects (also as header table in front of the original header). Fill with INSERT ... ACCEPTING DUPLICATE KEYS per package. Use NO_IMPORT for selection-only tables and generate for the source only; generate for source and target to migrate 1:1. Structural changes are detected when a run starts and trigger regeneration.	M5, SNP3
Global selection vs. own table	Global selection: one column, simple, list upload. Own table: dependent values, flexible, global rules apply, can be placed in the tree, documentable, works across scenarios, needs planning.	M5, SNP3
Global events	Effective for all tables unless the scope is limited to modules, objects or tables (Maintain Scope). Example: write company codes with object and table names into an own table. Another example calls SAVE_DATA_FOR_AT in the import event after package processing for Automated Testing.	SNP3
Global functions	Called with PERFORM from rules; must be activated for export and/or import (run parameters GLOBAL_FUNCTION_EXPORT / _IMPORT); interface of importing and changing parameters. Example: UMSCHL_NUMBER_GET_NEXT_CHAR calls DO_NRIV which calls NUMBER_GET_NEXT and maps its sy-subrc values to API_MESSAGE_ADD.	SNP3

Topic	Fact	Source
Standard global import rules	MANDT (client changed to target), LOGSYS (source logical system to target logical system, abort message if either is missing) and ADRNR (address numbers via global function).	SNP3
Offsets	Segment concatenated fields such as VAKEY by parameters (for example KVEWE, KOTABNR) and a segmentation string such as LIFNR(0,10),MATNR(10,18),EKORG(28,4),WERKS(32,4) for table A017. Relative position is typically 0 with no gaps; adjust when domain lengths change in a higher release (for example MATNR in S/4HANA).	SNP3
Mapping details	Generated mapping tables are /SNP/TE_; 1:1 entries are added automatically by API call. Standard content builds mappings as dependent mappings with the system alias as first source field. Pattern: API_MAPPING_GET first, ADD only if sy-subrc is not 0. Mapping vs. own table: a mapping has source and target values and is read by rule. Field rules override global rules and can read the whole record but change only their own field. Check CHECK NOT iv_value IS INITIAL first to save runtime.	M5, SNP2
Mapping tools	Bulk upload / download of mappings; validate values under Functions - Data Quality Mappings; report /SNP/TE05_OVERVIEW_TDEF_TABS lists generated tables. API_MAPPING_ADD dumps when the key exists, _SET overwrites, _CLEAR removes automatically added entries only.	M5, SNP2
Debugging	Dialog only, one process. Set BREAK-POINT or break user in the coding; step keys F5 into, F6 over, F7 out, F8 execute. Predefined debugging points exist for export and import events.	M5, SNP2

```
* Own table fill pattern (export) - /SNP/TE_AUFNR example from the training
* Data definition:         DATA: lt_te_aufnr TYPE TABLE OF /snp/te_aufnr, ls_te_aufnr TYPE /snp/te_aufnr.
* Before table processing: DELETE FROM /snp/te_aufnr WHERE scen_id = te-scen_id
*                            AND source = te-source_system_alias AND object_id = te-object_id
*                            AND table_id = te-table_alias.
* After data record:       ls_te_aufnr-aufnr = cs_aufk-aufnr.  APPEND ls_te_aufnr TO lt_te_aufnr.
* After package:           INSERT /snp/te_aufnr FROM TABLE lt_te_aufnr.
```

14.4 Number ranges and forerunners

Strategy	Behaviour	Needs
Increment (internal ranges in target)	Numbers are re-incremented in the target with NUMBER_GET_NEXT; mappings document old and new numbers.	Forerunners for a smooth run
Difference calculation (internal ranges)	New number = old number plus an NRIV offset; much faster, no forerunner or mapping. If no reserved number exists, NUMBER_GET_NEXT is used as fallback.	Save NRIV (several times for RBNZD), assign source systems, Update NRIV before import, buffer % and fixed buffer for RBNZD
1:1 (internal ranges)	Numbers adopted unchanged; number statuses updated in target.	-
Inactive, adopt 1:1 (external ranges)	External numbers adopted unchanged.	-

- **Forerunner** (a special data object, block type FOR): contains table references of existing objects, runs after the export and before the import, builds the mapping for key domains, has no export of its own and imports in simulation mode with one process per file. Only the mapping is created; nothing is written to the data tables. The linked objects must have been exported successfully. (M5, SNP3)
- **Why use it:** adjust exported data to the target (mappings for domains with internal number ranges), improve parallelization of the main import, and find errors early. Typical flow: export linked objects, import forerunner in simulation, then import the normal objects which read the built mapping. (M5, SNP3)
- **API_NUMBER_GET_NEXT** returns sy-subrc 0 when the value was calculated from the NRIV offset and 4 when it was generated with NUMBER_GET_NEXT; the strategy is returned too. After Update NRIV, the offsets are added to the old source numbers during import. (SNP3)
- **Simulation mode** runs the import without writing to the database and can be repeated as often as needed. (M5)

14.5 Execution control

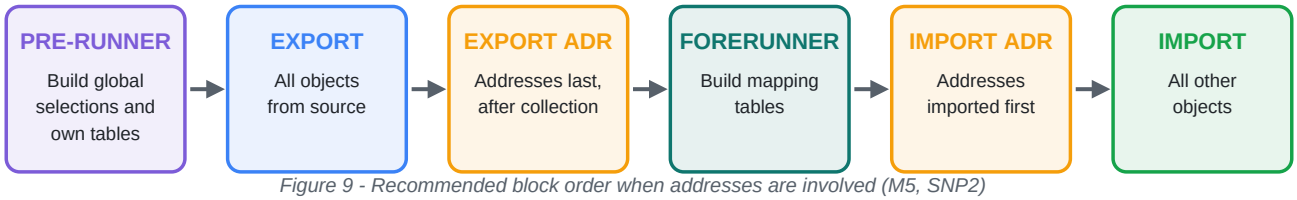


Figure 9 - Recommended block order when addresses are involved (M5, SNP2)

Topic	Fact	Source
Block types	EXP, FOR, IMP, UPL and DEL. Import blocks can be created from export / upload blocks; deletion blocks from import blocks after successful execution. Only active objects can be assigned; tables can be removed from an object inside a block without changing the object.	M5, SNP2
Address object	Address numbers are collected by global export rules for the domains ADRNR, AD_ADDRNUM, AD_PERSNUM, AD_SO_KEY and CADRNR. Export object ADR last and import it first.	SNP2
Dependencies	Predecessors order blocks and objects; missing dependencies cause problems or terminations. Multiple predecessors are allowed.	SNP2
Process control	Distributes processes by parameters such as table size. Max. processes above 0 on a block limits it and can prevent optimal control. For single executions 1 or 2 processes are recommended.	M5, SNP2
User priority	MIN runs at the end (1/10 of processes); empty = calculated or inherited (1/10); MEDIUM = calculated x50 (1/10); HIGH = calculated x100 (1/5); MAX runs first (1/2). Settable at block, object and table level and changeable during the transformation.	M5
Table locks	Lock set for cluster tables and tables with option Summarize (needed only for n:1 mapping); not set in DELETE mode. Switching the block mode adjusts locks on later tables. Disabling cluster locks saves runtime but is risky: with BSEG, several processes may write the same record into cluster RFBLG and cause a deadlock.	SNP3
Select export	Right-click the import block to assign exports manually, also from other systems of the scenario; with several source systems this is mandatory. Cross-system assignment can make target data inconsistent; tables that were not exported cannot be assigned.	SNP3, SNP2
Multiple satellite systems	Each system needs a unique alias (do not change it after global selections are maintained); a standard system is used where no system can be chosen (DDIC value help, syntax check). Global selection values are maintained per alias. Use environment variable te-source_system_alias in import coding for source-dependent mapping. Mappings added by API on one system are not transferred between systems. The system alias is set at block level.	SNP3
No shared file share	Report /SNP/TE02_FILE_TRANSFER moves export files from source share to target share via RFC through the control system; files are transferred only once, otherwise import ends with files-not-found. Transfer all files of a scenario together.	SNP3
Export-based customizing analysis	Analyzes customizing tables from an export. Green = filled identically, yellow = missing in source or target, red = filled with deviations in the target. Run it with all relevant objects and largely maintained organizational-unit selections, if needed in a separate scenario.	SNP3
RET scenario (save and restore)	Source = target. Save: export complete tables without selections or rules (the files are the backup). Reset: deletion object plus import block in mode DELETE. Restore: import the saved tables in mode INSERT without rule base, then repeat the original import with the corrected rule base.	SNP3
Deletion	Used for deletion projects (source and target at once) and for import correction. After deleting, clean up runs carefully because the transformation tree is the transformation history.	M5
Table statistics	Refresh in Systems after export programs are generated, after export forerunners and for filled own tables; the daemon refreshes SNP tables automatically (default interval 30 min).	SNP3

14.6 Validation and analysis tools

Tool	Fact	Source
Compare export to target	After import, compares the export file with the target table; key fields must be identical. Fields changed by rules (mapped, incremented) are expected differences.	SNP2
Test units	Stored reports and transactions called from the transformation in the target system.	SNP2
DCV prerequisites	Migration finished and check tables filled. Fields can be given a status to exclude irrelevant checks; tables a context (priority, responsibility, owner). Two checks from different times can be compared, for example before / after conversion or source / target.	SNP2
Automated Testing	Tests on application level (read mode, change mode) with random or selected data. With SAVE_DATA_FOR_AT the TC passes migrated data to /n/SNP/AT_mig_check on the target (requires the Migration CHECK add-on); the number tested can differ from the header table count.	SNP2, SNP3
Compare transformations	Compare runs of one or several scenarios by table (records and runtime) or by object (runtime); one leading run is the basis; CSV up and download.	M5, SNP2
Transformation log	/SNP/TCPROT; archived when tables are deleted or reset; Notes column for comments; progress shown as a percentage after each log refresh; duplicate keys shown during import or simulation, limited by LOG_MAX_ENTRIES.	M5, SNP2
Analyze DB indexes	In Data Objects - Functions, shows the secondary indexes with the best coverage of the selection fields (60% in the example) to help choose selections.	SNP3
PO Query	/n/SNP/CN_PO_QUERY queries cockpit persistent objects (classes /SNP/TE01_CL_SCENARIO, _MODULE, _OBJECT, _TABLE); can show change information and history. Start simple and refine; default maximum 500 records.	SNP3
Change history	Technical log of changes in data objects, global and systems, with user and time. SAP menu - Information - Change history.	M5
Back in Mission Control	Final training steps: copy template runs to executable runs, execute and verify runs, run the final syntax check, check that the runs exist in TC.	M5

Reference: Kyano Mission Control Complete Reference Guide - chapters 1-7 Mission Control, 10 Transformation Cockpit, 11 Transformation Cockpit Training Guide (v19.03), 12 CrystalBridge Analysis (release 20.07); CrystalBridge Transformation User Guide 25.09 SP00; Module 5 Transformation Advanced (v25.03); Transformation Cockpit SNP2 and Advanced SNP3 training decks (19.03 SP00). SNP Support: support.snpgroup.com, support-tc@snpgroup.com. Document generated with IBM Bob.